

LITERATURE SURVEY

Airline & Airport Operations – Flight Booking Management & Analytics Portal

The literature survey reviews research related to airline seat inventory, booking and revenue management, customer satisfaction, and flight-delay analytics. The purpose is to understand existing approaches and identify the gap addressed by the proposed project.

S.No.	Author / Year	Research Area	Method / Technique	Key Finding	Limitation / Research Gap
1	Hung & Chen (2013)	Airline seat allocation	Dynamic seat rationing and expected revenue-gap approaches	Studied decisions for accepting or rejecting booking requests while managing limited seats.	Focuses on revenue-management decisions rather than a complete booking and analytics portal.
2	de Boer, Freling & Piersma (2002)	Airline network revenue management	Mathematical programming and simulation	Investigated booking limits and seat-inventory control for airline networks.	Primarily an optimization/revenue-management model rather than an integrated web-based system.
3	Lin et al. (2020)	Airline seat inventory control	Hybrid static/dynamic model and stochastic dynamic programming	Proposed a hybrid model for multi-leg airline seat inventory management.	Concentrates on seat-inventory optimization; user booking and operational dashboards are outside the main scope.
4	Hung et al. (2021)	Airline group booking	Analytical booking model / bid-price control	Studied booking policies considering group arrivals, cancellations and refunds.	Focuses on group-booking behavior rather than general passenger booking and airport analytics.
5	Dolnicar et al. (2008)	Airline revenue management & customer satisfaction	Regression and ANOVA analysis	Examined how seat-inventory control and denied boarding can affect customer satisfaction.	Focuses on customer response rather than implementing a complete booking-management system.
6	Zhong et al. (2025)	Flight-delay analytics	Machine-learning-based prediction and optimization	Investigated prediction and optimization of civil-aviation flight delays.	Concentrates on delay prediction; it does not combine booking, seat management and analytics into one portal.

Research Gap

Existing studies mainly focus on individual airline problems such as seat-inventory control, revenue management, group booking, customer satisfaction, or flight-delay prediction. The proposed project addresses an integration gap by bringing flight booking, passenger/flight/airport database management, seat availability, and operational analytics together in a single database-driven portal.

Relevance to Our Project

The survey supports the major components of the project: booking management, seat availability, route and booking analytics, and delay-related analysis. The project extends these themes into one system using a frontend, Python/Flask backend, and MySQL database.

References

Hung, C.-W., & Chen, C.-H. (2013). Airline seat allocation / dynamic seat rationing research.

de Boer, E. R., Freling, R., & Piersma, N. (2002). Mathematical programming and simulation approaches to airline network revenue management.

Lin et al. (2020). Research on hybrid static/dynamic airline seat-inventory control and stochastic dynamic programming.

Hung et al. (2021). Research on airline group booking and bid-price control.

Dolnicar et al. (2008). Research examining airline revenue management, denied boarding and customer satisfaction.

Zhong et al. (2025). Research on civil-aviation flight-delay prediction and optimization.